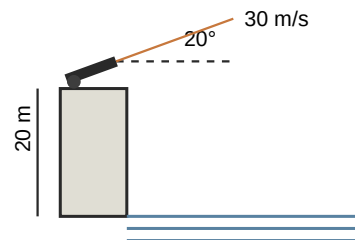


Projectile motion: two motions at once

Name _____ Date _____ Take $g = 9.8 \text{ m/s}^2$. Ignore air resistance.

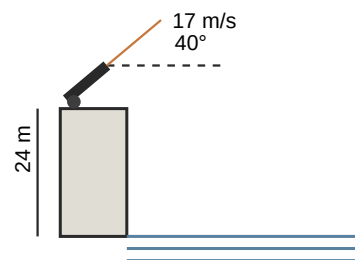
1. A cannon fires at 30 m/s , 20° above the horizontal, from 20 m above the sea. What is the horizontal component of the launch velocity?

Answer (m/s): _____



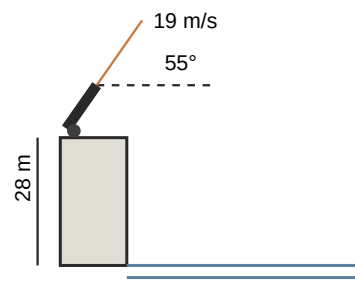
2. A cannon fires at 17 m/s , 40° above the horizontal, from 24 m above the sea. How long is the ball in the air before it hits the water?

Answer (s): _____



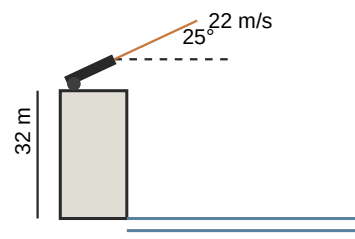
3. A cannon fires at 19 m/s , 55° above the horizontal, from 28 m above the sea. How far from the cannon does the ball hit the water?

Answer (m): _____



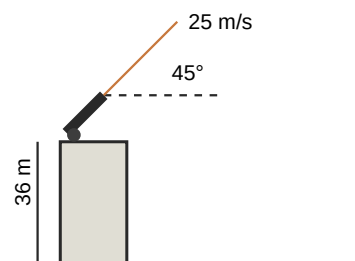
4. A cannon fires at 22 m/s , 25° above the horizontal, from 32 m above the sea. What is the greatest height of the ball above the sea?

Answer (m): _____



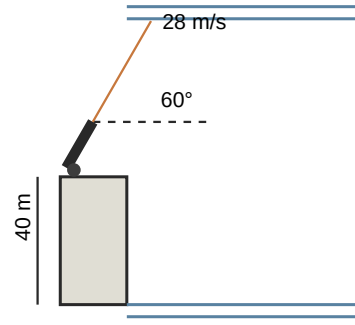
5. A cannon fires at 25 m/s , 45° above the horizontal, from 36 m above the sea. What is the horizontal component of the launch velocity?

Answer (m/s): _____



6. A cannon fires at 28 m/s , 60° above the horizontal, from 40 m above the sea. How long is the ball in the air before it hits the water?

Answer (s): _____



Answer key

1. 28.2 m/s

$$v_x = v_0 \cos \text{angle} \quad > \quad v_x = 30 \times \cos 20^\circ \quad > \quad v_x = 28.2 \text{ m/s}$$

2. 3.59 s

$$0 = h + v_y t - \frac{1}{2} g t^2 \text{ with } v_y = 10.9 \quad > \quad 4.9 t^2 - 10.9 t - 24 = 0 \quad > \quad t = (10.9 + \sqrt{10.9^2 + 2 \times 9.8 \times 24}) / (9.8) \quad > \quad t = 3.59 \text{ s}$$

3. 48.5 m

$$t = 4.45 \text{ s (time of flight)} \quad > \quad x = v_x t = 10.9 \times 4.45 \quad > \quad x = 48.5 \text{ m}$$

4. 36.4 m

$$\text{At the top } v_y = 0: v_y^2 = 2 g \text{ change in } y \quad > \quad \text{change in } y = (9.3^2) / (2 \times 9.8) = 4.4 \text{ m} \quad > \quad y_{\text{max}} = 32 + 4.4 = 36.4 \text{ m}$$

5. 17.7 m/s

$$v_x = v_0 \cos \text{angle} \quad > \quad v_x = 25 \times \cos 45^\circ \quad > \quad v_x = 17.7 \text{ m/s}$$

6. 6.25 s

$$0 = h + v_y t - \frac{1}{2} g t^2 \text{ with } v_y = 24.2 \quad > \quad 4.9 t^2 - 24.2 t - 40 = 0 \quad > \quad t = (24.2 + \sqrt{24.2^2 + 2 \times 9.8 \times 40}) / (9.8) \quad > \quad t = 6.25 \text{ s}$$